



CS 324:
Modeling and Simulation

**Random-Number
Generation**

Chapter 5: Ran. Num. Gen.

- Properties of Random Numbers.
- Generation of Pseudo-Random Numbers.
- Techniques for Generating Pseudo-Random Numbers.
- Tests for Random Numbers.

Tests for PRNs (1/3)

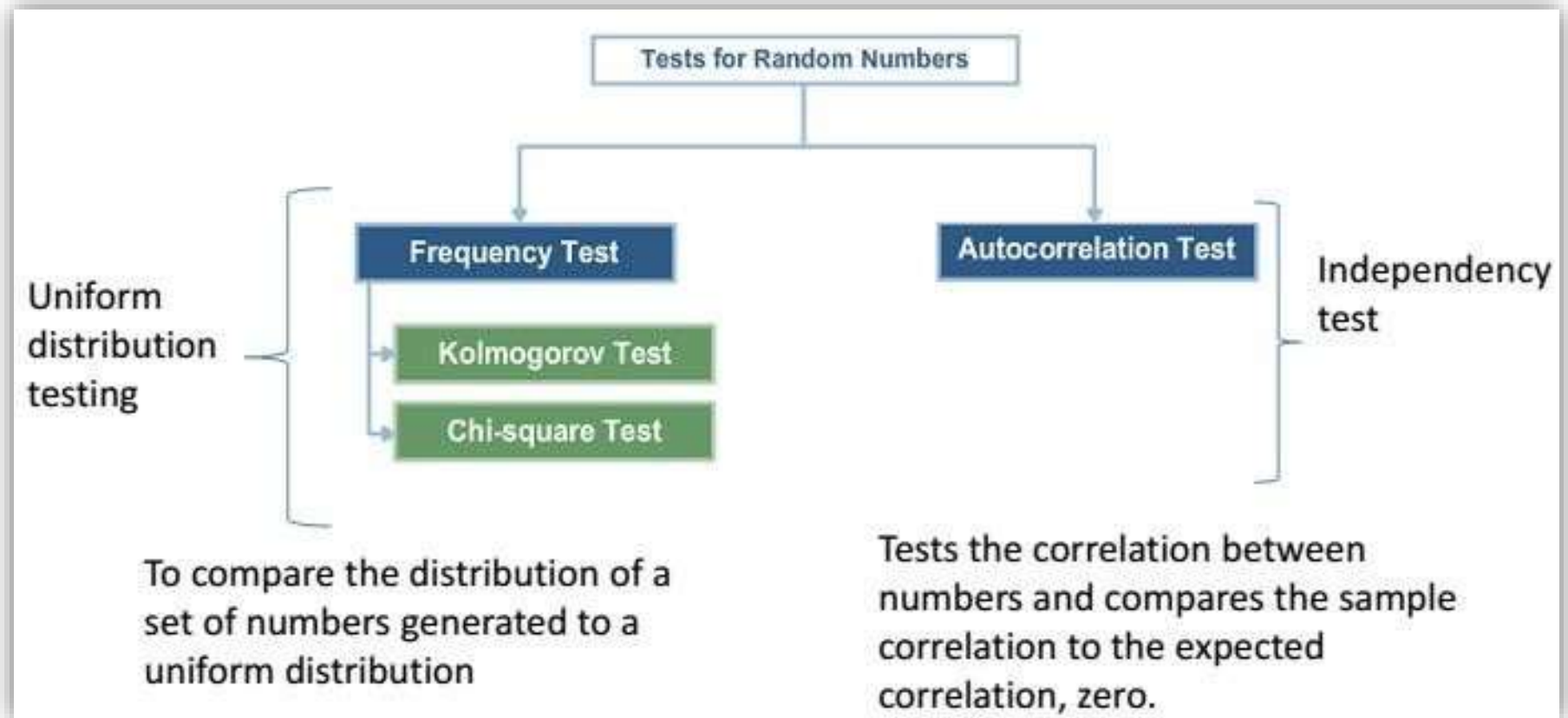
Introduction (1/5)

The properties of random numbers: Uniformity and Independence. Number of tests can be performed to check these properties been achieved or not.

- **Frequency Test:** Uses the Kolmogorov-Smirnov or the Chi-square test to compare the distribution of the set of numbers generated to a uniform distribution.
- **Autocorrelation Test:** Tests the correlation between numbers and compares the sample correlation to the expected correlation, zero.

Tests for PRNs (1/3)

Introduction (2/5)



Tests for PRNs (1/3)

Introduction (3/5)

In testing for uniformity, the hypotheses are as follows:

$$H_0: R_i \sim \text{Uniform}[0, 1]$$

$$H_1: R_i \not\sim \text{Uniform}[0, 1]$$

The null hypothesis, H_0 , reads that the numbers are distributed uniformly on the interval $[0, 1]$. Failure to reject the null hypothesis means that evidence of nonuniformity has not been detected by this test. This does not imply that further testing of the generator for uniformity is unnecessary.

Tests for PRNs (1/3)

Introduction (4/5)

In testing for independence, the hypotheses are as follows:

$$H_0: R_i \sim \text{independently}$$

$$H_1: R_i \not\sim \text{independently}$$

This null hypothesis, H_0 , reads that the numbers are independent. Failure to reject the null hypothesis means that evidence of dependence has not been detected by this test. This does not imply that further testing of the generator for independence is unnecessary.

Tests for PRNs (1/3)

Introduction (5/5)

For each test, a level of significance α must be stated. The level α is the probability of rejecting the null hypothesis when the null hypothesis is true:

$$\alpha = P(\text{reject } H_0 | H_0 \text{ true})$$

The decision maker sets the value of α for any test.

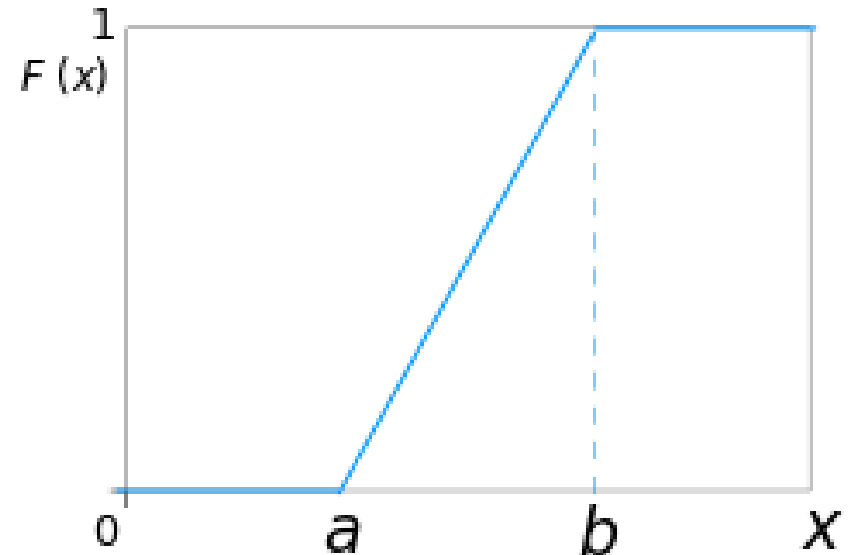
Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (1/5)

Uniformity

Compares the continuous cumulative distribution function (cdf), $F(x)$, of the uniform distribution with the empirical distribution fun. $S_N(x)$ of the sample of N observations.

$$F(x) = \begin{cases} 0, & x \leq a \\ \frac{x - a}{b - a}, & a < x < b \\ 1, & x \geq b \end{cases}$$



Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (1/5)

Uniformity

Compares the continuous cumulative distribution function (cdf), $F(x)$, of the uniform distribution with the empirical distribution fun. $S_N(x)$ of the sample of N observations.

$$D = \max|F(x) - S_N(x)|$$

The sampling distribution of D is known; it is tabulated as a function of N in the following Table.

Tests for PRNs (2/3)

Kolmogorov--Smirnov Critical Values

<i>Degrees of Freedom</i> (N)	$D_{0.10}$	$D_{0.05}$	$D_{0.01}$
1	0.950	0.975	0.995
2	0.776	0.842	0.929
3	0.642	0.708	0.828
4	0.564	0.624	0.733
5	0.510	0.565	0.669
6	0.470	0.521	0.618
7	0.438	0.486	0.577
8	0.411	0.457	0.543
9	0.388	0.432	0.514
10	0.368	0.410	0.490

11	0.352	0.391	0.468
12	0.338	0.375	0.450
13	0.325	0.361	0.433
14	0.314	0.349	0.418
15	0.304	0.338	0.404
16	0.295	0.328	0.392
17	0.286	0.318	0.381
18	0.278	0.309	0.371
19	0.272	0.301	0.363
20	0.264	0.294	0.356
25	0.24	0.27	0.32
30	0.22	0.24	0.29
35	0.21	0.23	0.27
Over 35	$\frac{1.22}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.63}{\sqrt{N}}$

Tests for PRNs (2/3)

Kolmogorov--Smirnov Critical Values

Degrees of Freedom (N)	$D_{0.10}$	$D_{0.05}$	$D_{0.01}$
1	0.950	0.975	0.995
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9	0.388	0.432	0.514
10	0.368	0.410	0.490

if $N = 8$ and $\alpha = 0.05$
 $D_{0.05} = 0.457$

12	0.358	0.375	0.391
13	0.325	0.361	0.433
14	0.314	0.349	0.418
15	0.304	0.338	0.404
16	0.295	0.328	0.392
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35	0.21	0.23	0.27
Over 35	$\frac{1.22}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.63}{\sqrt{N}}$

N	0.01	0.05	0.1	0.15	0.2
-	0.995	0.975	0.950	0.925	0.900
2	0.929	0.842	0.776	0.726	0.684
3	0.828	0.708	0.642	0.597	0.565
4	0.733	0.624	0.564	0.525	0.494
5	0.669	0.565	0.510	0.474	0.446
6	0.618	0.521	0.470	0.436	0.410
7	0.577	0.486	0.438	0.405	0.381
8	0.543	0.457	0.411	0.381	0.358
9	0.514	0.432	0.388	0.360	0.339
10	0.490	0.410	0.368	0.342	0.322
11	0.468	0.391	0.352	0.326	0.307
12	0.450	0.375	0.338	0.313	0.295
13	0.433	0.361	0.325	0.302	0.284
14	0.418	0.349	0.314	0.292	0.274
15	0.404	0.338	0.304	0.283	0.266
16	0.392	0.328	0.295	0.274	0.258
17	0.381	0.318	0.286	0.266	0.250
18	0.371	0.309	0.278	0.259	0.244
19	0.363	0.301	0.272	0.252	0.237
20	0.356	0.294	0.264	0.246	0.231
25	0.320	0.270	0.240	0.220	0.210
30	0.290	0.240	0.220	0.200	0.190
35	0.270	0.230	0.210	0.190	0.180
40	0.250	0.210	0.190	0.180	0.170
45	0.240	0.200	0.180	0.170	0.160
50	0.230	0.190	0.170	0.160	0.150
OVER 50	1.63	1.36	1.22	1.14	1.07
	$\frac{1.63}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.22}{\sqrt{N}}$	$\frac{1.14}{\sqrt{N}}$	$\frac{1.07}{\sqrt{N}}$

Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (2/5)

Uniformity

For testing against a uniform cdf, the test procedure follows these steps:

Step1: Rank the data from smallest to largest. Let R_i denote i^{th} smallest observation, so that $R_1 \leq R_2 \leq \dots R_N$

Step2: Null hypothesis $H_0: R'_i \sim U(0,1)$

Distributed as uniform distribution

Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (3/5)

Uniformity

Step3: Compute D^+ and D^- , where

$$D^+ = \max \left\{ \frac{i}{N} - R_i \right\}, \quad D^- = \max \left\{ R_i - \frac{i-1}{N} \right\}$$

Step4: Compute $D = \max\{D^+, D^-\}$

Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (3/5)

Uniformity

Step3: Compute D^+ and D^- , where

$i = 1, 2, \dots, N$

$$D^+ = \max \left\{ \frac{i}{N} - R_i \right\}, \quad D^- = \max \left\{ R_i - \frac{i-1}{N} \right\}$$

Step4: Compute $D = \max\{D^+, D^-\}$

Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (4/5)

Uniformity

Step5: Locate the critical value D_α in the K-S Table for the specified significance level α and the sample size N .

Kolmogorov--Smirnov Critical Values

Degrees of Freedom (N)	$D_{0.10}$	$D_{0.05}$	$D_{0.01}$
1	0.950	0.975	0.995
2	0.776	0.842	0.929
3	0.642	0.708	0.828
4	0.564	0.624	0.733
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35	0.21	0.23	0.27
Over 35	$\frac{1.22}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.63}{\sqrt{N}}$

Tests for PRNs (2/3)

Kolmogorov-Smirnov (K-S) Test (5/5)

Uniformity

Step6: If the sample statistic D is *greater than* the critical value D_α , the null hypothesis is rejected. If $D \leq D_\alpha$, conclude that we fail to reject the null hypothesis and these PRNs can be accepted according to K-S test.

Tests for PRNs (3/3)

Example – (K-S) Test (1/5)

Uniformity

Suppose 5 generated numbers are 0.44, 0.81, 0.14, 0.05, 0.93.

For the specified significance level $\alpha = 0.05$.

Tests for PRNs (3/3)

Example – (K-S) Test (1/5)

Uniformity

Suppose 5 generated numbers are 0.44, 0.81, 0.14, 0.05, 0.93.

For the specified significance level $\alpha = 0.05$.

$$N = 5 \text{ and } \alpha = 0.05$$

Tests for PRNs (3/7)

Example – (K-S) Test (2/5)

Uniformity

$N = 5$ and $\alpha = 0.05$
 $D_{0.05} = 0.565$

Degrees of Freedom (N)	$D_{0.10}$	$D_{0.05}$	$D_{0.01}$
1	0.950	0.975	0.995
2	0.776	0.842	0.929
3	0.642	0.708	0.828
4	0.564	0.624	0.733
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Tests for PRNs (3/3)

Example – (K-S) Test (3/5)

Uniformity

The 5 generated numbers are 0.44, 0.81, 0.14, 0.05, 0.93.

First, the numbers must be **sorted** from smallest to largest.

R_i	0.05	0.14	0.44	0.81	0.93

Tests for PRNs (3/7)

Example – (K-S) Test (3/5)

Uniformity

$N=5$

R_i	0.05	0.14	0.44	0.81	0.93

Tests for PRNs (3/3)

Example – (K-S) Test (3/5)

Uniformity

$N=5$

R_i	0.05	0.14	0.44	0.81	0.93

Tests for PRNs (3/7)

Example – (K-S) Test (3/5)

Uniformity

$N=5$

R_i	0.05	0.14	0.44	0.81	0.93
i/N	0.20	0.40	0.60	0.80	1.00

Tests for PRNs (3/3)

Example – (K-S) Test (3/5)

Uniformity

$N=5$

R_i	0.05	0.14	0.44	0.81	0.93
i/N	0.20	0.40	0.60	0.80	1.00
$i/N - R_i$	0.15	0.26	0.16	–	0.07

$D^+ = 0.26$

Tests for PRNs (3/7)

Example – (K-S) Test (3/5)

Uniformity

$N=5$

R_i	0.05	0.14	0.44	0.81	0.93
i/N	0.20	0.40	0.60	0.80	1.00
$i/N - R_i$	0.15	0.26	0.16	—	0.07
$(i - 1)/N$	0.00	0.20	0.40	0.60	0.80

$D^+ = 0.26$

Tests for PRNs (3/3)

Example – (K-S) Test (3/5)

Uniformity

$N=5$

R_i	0.05	0.14	0.44	0.81	0.93
i/N	0.20	0.40	0.60	0.80	1.00
$i/N - R_i$	0.15	0.26	0.16	—	0.07
$(i - 1)/N$	0.00	0.20	0.40	0.60	0.80
$R_i - (i - 1)/N$	0.05	—	0.04	0.21	0.13

$$D^+ = 0.26$$

$$D^- = 0.21$$

Tests for PRNs (3/7)

Example – (K-S) Test (4/5)

Uniformity

$$D = \max\{D^+, D^-\} = 0.26$$

$$D^+ = 0.26$$

$$D^- = 0.21$$

Tests for PRNs (3/3)

Example – (K-S) Test (5/5)

Uniformity

Since the computed value, 0.26, is less than the critical value, 0.565, the hypothesis that the distribution of the generated numbers is the uniform distribution is “*failed to reject*”.

$$D = \max\{D^+, D^-\} = 0.26 < D_\alpha$$

$$D_\alpha = 0.565$$

Tests for PRNs (3/3)

Example – (K-S) Test (1/5)

Uniformity

Suppose 5 generated numbers are:
0.01, 0.02, 0.03, 0.04, 0.99

Significance level: $\alpha = 0.05$

Thank You!